MPH 小型化金属化聚丙烯膜电容器(浸渍型) 105°C

MPH Mini-sized Metallized Polypropylene Film Capacitor (Dipped) 105°C



系列 Series MPH

行业型号 Industry code C37

产品类型 Type

小型化金属化聚丙烯膜电容器(浸渍型)
Mini-sized Metallized Polypropylene Film Capacitor (Dipped)

应用 Applications

- 用于开关电源，电子镇流器和变频器等中间电路直流滤波(DC-link, PFC 等)
  As intermediate circuit capacitor for SMPS, Electronic Ballast inverter (i.e. DC-link, DC filter and P.F.C)

技术要求 Specifications

项目 Item	参数 Specification
引用标准 Reference Standard	GB/T 10190 (IEC 60384-16)
气候类别 Climatic Category	40/105/21
工作温度范围 Operating Temperature Range	-40°C~+105°C (+85°C~+105°C): derating factor 1.5% per °C for R.V(DC)
额定电压 Rated voltage	400VDC/450VDC, 500VDC/550VDC, 630VDC
容量范围 Capacitance range	0.01μF~2.2μF
容差 Capacitance tolerance	±5% (J), ±10% (K)
耐电压 Voltage Proof	1.6* R.V(DC), 2s (between terminals)
损失角 Dissipation Factor	≤0.1% (1kHz at 20~25°C)
绝缘阻抗 Insulation Resistance	≥30 000MΩ, C_N≤0.33μF; RC_N≥10 000 s, C_N>0.33μF (20°C, 100V, 1min)

使用注意 Usage Note

注：MPH 产品只适用于 DC 滤波和 DC 模块电路，意指施加给电容的电压必须是单向涟波电压，电压代表波形如下图示。不适用于交流电路，对此有何疑问请和我们的技术人员联系。

a: The MPH series is only recommended to use in DC-filter or DC-blocking circuits. It means the voltage applied to the capacitors must be unidirectional ripple voltage. The typical voltage curve is as following reference. If you have any questions for this note, please feel free to contact our technical engineer.

Here: V1≥0, V2≤U_R, Vrms=(V2−V1)÷√2, Irms=2πf×C×(V1−V2)÷√2; U_R is the rated voltage of the capacitor

最大脉冲爬升速率 Maximum Pulse Rise Time

最大脉冲爬升速率 dV/dt (V/μs) Maximum Pulse Rise Time (dV/dt)

U_R (V)	P=7.5	P=10	P=15	P=22.5
400/450	320	300	200	100
500/550	380	350	220	150
630	420	400	300	180

若实际工作电压 U 比额定电压 U_R 低，电容器可工作在更高的 dV/dt 场合。这样 dV/dt 允许值应为右表值乘 U_R/U。
If the working voltage (U) is lower than the rated voltage (U_R), the capacitor can be worked at a higher dV/dt. In this case, the maximum allowed dV/dt is obtained by multiplying the right value with U_R/U.

外形尺寸 Dimensions

外形尺寸 450VDC Dimensions (mm) 450VDC

C_R (μF)	W (±1)	H (±1)	T (±1)	P (±1)	DΦ (±0.05)
0.01	10.5	9	6	7.5	0.6
0.022	10.5	9	6	7.5	0.6
0.033	10.5	9.5	6.5	7.5	0.6
0.047	10.5	12.5	7	7.5	0.6
0.068	10.5	13.5	8	7.5	0.6
0.1	10.5	14.5	9	7.5	0.6
0.01	12.5	9	6	10	0.6
0.022	12.5	9	6	10	0.6
0.033	12.5	9	6	10	0.6
0.047	12.5	11	7	10	0.6
0.068	12.5	12	8	10	0.6
0.1	12.5	10	6	10	0.6
0.15	12.5	11	6.5	10	0.6
0.22	12.5	13.5	7	10	0.6
0.33	12.5	14	8	10	0.6
0.47	12.5	16.5	8	10	0.6
0.68	12.5	19	10	10	0.8
1.0	12.5	23	10	10	0.8
0.1	17.5	13.5	8	15	0.8
0.22	17.5	14	7.5	15	0.8
0.33	17.5	13	9	15	0.8
0.47	17.5	14	6.5	15	0.6
0.68	17.5	15	8	15	0.8
1.0	17.5	17	8.5	15	0.8
1.5	17.5	23.5	10	15	0.8
2.2	17.5	25.5	11.5	15	0.8
1.5	25.5	17.5	9	22.5	0.8
2.2	25.5	20	10	22.5	0.8

外形尺寸 550VDC Dimensions (mm) 550VDC

C_R (μF)	W (±1)	H (±1)	T (±1)	P (±1)	DΦ (±0.05)
0.01	10.5	9	6	7.5	0.6
0.022	10.5	9	6	7.5	0.6
0.033	10.5	9.5	6.5	7.5	0.6

C_R (μF)	W (±1)	H (±1)	T (±1)	P (±1)	DΦ (±0.05)
0.047	10.5	12.5	7	7.5	0.6
0.01	12.5	10	6	10	0.6
0.015	12.5	10	6	10	0.6
0.022	12.5	10	6	10	0.6
0.033	12.5	10	6.5	10	0.6
0.047	12.5	11	7.5	10	0.6
0.068	12.5	12	8	10	0.6
0.1	12.5	10	7	10	0.6
0.15	12.5	12	7	10	0.6
0.22	12.5	14	8	10	0.6
0.1	17.5	13.5	8	15	0.8
0.15	17.5	13.5	8	15	0.8
0.22	17.5	13.5	8	15	0.8
0.33	17.5	14	8.5	15	0.8
0.47	17.5	15	9.5	15	0.8
0.68	17.5	17	10.5	15	0.8
1.0	17.5	20	12	15	0.8
0.68	25.5	16.5	10	22.5	0.8
1.0	25.5	21	11	22.5	0.8
1.5	25.5	23.5	13.5	22.5	0.8
2.2	25.5	26.5	16.5	22.5	0.8

外形尺寸 630VDC Dimensions (mm) 630VDC

C_R (μF)	W (±1)	H (±1)	T (±1)	P (±1)	DΦ (±0.05)
0.01	10.5	9	5.5	7.5	0.6
0.022	10.5	12.5	7	7.5	0.6
0.033	10.5	14	8	7.5	0.6
0.01	12.5	10	6	10	0.6
0.015	12.5	10	6	10	0.6
0.022	12.5	10	6	10	0.6
0.033	12.5	10	6.5	10	0.6
0.047	12.5	11	7	10	0.6
0.068	12.5	11	8	10	0.6
0.1	12.5	15	7	10	0.6
0.22	12.5	15.5	10.5	10	0.6
0.1	17.5	13.5	8	15	0.8
0.22	17.5	13.5	8	15	0.8
0.33	17.5	15.5	9.5	15	0.8
0.39	17.5	17.5	9.5	15	0.8
0.47	17.5	18	10	15	0.8
0.68	17.5	21	11.5	15	0.8
1.0	17.5	25	15	15	0.8
0.68	25.5	17	10	22.5	0.8
1.0	25.5	21	11	22.5	0.8
1.5	25.5	23.5	13.5	22.5	0.8
2.2	25.5	26.5	16.5	22.5	0.8

备注 Notes

- **产品标识示例 Marking example:** BKW MPH / 824K400V